EmacEArt Skybox URP

Procedural sky shader for Unity Universal Render Pipeline - user manual

What this is

A fully procedural sky for Unity URP. The whole dome is generated from math in the fragment shader: gradient, sun disc and glow, atmospheric haze, animated clouds and a star field. No panorama texture, no cubemap, no post-processing volume is needed for the sky itself. Every control is a material property, so a scene can go from clear noon to overcast dusk by dragging sliders, and the result is visible in the Scene view immediately.

What you get

- **Three-zone gradient** - zenith, horizon and ground colours blend across the dome, with sharpness and vertical offset of the boundary.
- **Procedural sun** - a crisp disc with a separate glow halo. It follows the scene Main Directional Light by default, so rotating the light moves the sun.
- **Atmospheric haze** - a tinted band around the horizon for aerial perspective, from a clear day to a thick misty morning.
- **Azimuthal gradient** - a directional colour sweep across the sky, aimed by compass angle. Sunsets and golden hours come from this one.
- **Animated clouds, five styles** - Standard, Feathered, Round, Diagonal and Cotton, with coverage, density, softness, stretch, swirl, banding and scroll speed.
- **Star field** - cubemap based, with intensity, threshold and sharpness, switched on and off independently from the clouds.
- **Built-in colour grading** - exposure, contrast, saturation and full lift / gamma / gain applied to the sky output only.

Requirements

Item	Value
Unity	2022.3 LTS or newer
Render pipeline	Universal Render Pipeline (URP). The shader includes the URP shader library, so it does not compile in Built-in or HDRP.
Platforms	Everything URP itself supports. No compute shaders, no render textures, no scripts at runtime.
Runtime code	None. The only C# file is an editor-only inspector and it never ships into a player build.

Installation

- Import the package into your project. Everything lands under **Assets/EmacEArt/SkyboxURP**.
- Open **Assets/EmacEArt/SkyboxURP/Scenes/EA_SkyboxURP_Demo.unity** to see the sky with its default settings.
- To use it in your own scene: **Window > Rendering > Lighting > Environment**, and set **Skybox Material** to **EA_Skybox**.
- Duplicate **EA_Skybox** before you start tuning, so the shipped defaults stay available as a reference to go back to.

Ambient light from the sky

In the Lighting window set **Environment Lighting > Source** to **Skybox** and press **Generate Lighting** (or leave Auto Generate on). The ambient probe is then rebuilt from this sky, so a warm sunset sky warms up

everything standing in the scene. The demo scene is already set that way.

Inspector reference

Select the material and the custom inspector groups every property into the seven sections below. The Default column is the built-in default of each control. The material that ships with the package is a tuned daytime sky, so some of its values differ; the table right below lists exactly which.

What the shipped material changes

Control	Shipped value	Why
Zenith / Horizon Color	0.13, 0.32, 0.72 / 0.62, 0.78, 0.92	Daylight blue overhead fading to a pale band at eye level.
Horizon Sharpness	2.6	A wide soft blend instead of a tight band.
Atmosphere	tint 0.55, 0.70, 0.92 / strength 0.34 / falloff 4.0	Light haze that stays near the horizon line.
Sky Gradient	colour 1.0, 0.62, 0.30 / angle 330 / spread 0.62 / strength 0.30	A warm sweep aimed at the sun side of the sky.
Sun	size 0.006 / glow size 0.28 / falloff 6.5 / glow 1.0, 0.85, 0.62	A small disc with a wide warm halo around it.
Clouds	on, style Cotton, coverage 0.54, scale 2.4, softness 0.09, stretch 0.36, swirl 0.18, speed 0.018	Lobed drifting masses rather than an even overcast.
Stars	off	It is a daytime sky.
Exposure / Saturation	1.08 / 1.12	A small lift so the blue does not go flat.

Zenith and Horizon

Property	Range	Default	What it does
Zenith Color	colour	0.05, 0.08, 0.22	Colour straight overhead.
Horizon Color	colour	0.38, 0.55, 0.82	Colour at eye level.
Ground Color	colour	0.12, 0.10, 0.08	Colour of the lower hemisphere, below the horizon line.
Horizon Sharpness	0.5 - 16	4.0	How fast horizon turns into zenith. Low is a wide soft blend, high presses the colour change into a narrow band.
Horizon Offset	-0.5 - 0.5	0.0	Moves the gradient up or down without moving the ground hemisphere itself.

Sun

Property	Range	Default	What it does
Sun Follows Main Light	on / off	on	Takes the direction from the scene Main Directional Light every time the inspector refreshes. Turn it off to aim the sun by hand.
Sun Direction	vector	0.3, 0.6, 0.5	Manual direction. Read-only while the toggle above is on.
Sun Color	colour	white	Colour of the disc itself.
Sun Glow Color	colour	white	Colour of the halo around the disc.
Sun Size	0.0005 - 0.05	0.003	Angular size of the disc. The real sun sits near the low end of this range.
Sun Glow Size	0.01 - 1.0	0.132	How far the halo reaches out from the disc.
Sun Glow Falloff	1 - 16	16.0	Steepness of the halo. High values keep the glow tight, low values spread a wide bloom across the sky.

Sky Gradient (azimuthal)

A second gradient that runs sideways instead of vertically, aimed like a compass. Strength is the master switch: zero by default, 0.30 in the shipped material.

Property	Range	Default	What it does
Gradient Color	colour	1.0, 0.45, 0.10	The tint swept across the sky.
Direction (deg)	0 - 360	0	Compass bearing the tint comes from.
Spread	0.01 - 1.0	0.5	Falloff width. Near 1 the tint covers the whole sky, near 0.1 it stops at a hard half-sphere.
Strength	0.0 - 1.0	0.0	Master amount. Raise it to switch the effect on.

Atmosphere (aerial perspective)

Property	Range	Default	What it does
Atmosphere Tint	colour	0.20, 0.35, 0.60	Colour of the haze band added around the horizon.
Atmosphere Strength	0.0 - 1.5	0.55	How much haze is added. Above roughly 1.0 the horizon starts to glow.
Atmosphere Falloff	1 - 10	2.5	How quickly the haze fades with height. High values pin it to the horizon line.

Color Grading

Applied to the sky output only, after everything else. Nothing else in the scene is touched, so this does not replace a post-processing volume.

Property	Range	Default	What it does
Exposure	0.1 - 4.0	1.0	Overall brightness multiplier.
Contrast	0.5 - 2.0	1.05	Spread around mid grey.
Saturation	0.0 - 2.5	1.1	Colour strength. Zero gives a grey sky.
Lift (Shadows)	colour	0, 0, 0	Adds to the dark end.
Gamma (Midtones)	colour	0.5, 0.5, 0.5	Neutral point is 0.5. Higher brightens the midtones, lower darkens them.
Gain (Highlights)	colour	1, 1, 1	Multiplies the bright end.

Stars

Stars need a cubemap. The shipped daytime material has them switched off and the slot empty. For a night sky switch them on and assign a cubemap; any star cubemap works.

Property	Range	Default	What it does
Stars Enabled	on / off	on	Shader keyword. Off removes the whole branch from the compiled shader.
Stars Cubemap	cubemap	empty	Source texture. Brightness of each texel decides whether a star appears there.
Stars Intensity	0.0 - 4.0	1.0	Brightness of the stars that pass the threshold.
Stars Threshold	0.0 - 1.0	0.72	Cut-off. Lower it for a denser field, raise it to keep only the brightest points.
Stars Sharpness	1 - 32	8.0	How abrupt the cut-off is. High values give hard pinpoints.

Clouds

Off by default, on in the shipped material. Switching Clouds Enabled on compiles the cloud branch and starts the animation; the layer scrolls on its own in the Scene view.

Property	Range	Default	What it does
Clouds Enabled	on / off	off	Shader keyword. Off costs nothing at all.
Cloud Color	colour	white	Colour of the lit part of a cloud.
Cloud Shadow	colour	0.72, 0.78, 0.88	Colour of the shaded part. The two mix by band level.
Coverage	0.0 - 1.0	0.5	How much of the sky is covered. The edge creeps outward smoothly as you drag.
Softness	0.001 - 0.5	0.15	Width of the fade at a cloud edge. Low values give a cut-out silhouette.
Bands (posterize)	1 - 8	1.0	Number of flat steps. One is smooth; two and up give a stepped, cel-shaded look.
Density	0.0 - 1.0	1.0	Opacity of the layer over the sky behind it.
Stretch	0.0 - 0.97	0.3	Horizontal smearing through a domain warp. The effect is squared, so it stays gentle for most of the slider and goes dramatic at the top.
Swirl	0.0 - 1.0	0.0	Second warp with a rotating component. Turns straight streaks into curling masses.
Style	0 - 4	0 (Standard)	Pattern generator: 0 Standard, 1 Feathered (cirrus ridges), 2 Round (separated blobs), 3 Diagonal, 4 Cotton (merged lobed masses).
Scale	0.5 - 8.0	3.0	Size of the pattern. High values make many small clouds.
Min Elevation	0.0 - 0.5	0.05	Height above the horizon where clouds are allowed to start. Raise it to keep the horizon clear.
Speed	0.0 - 1.0	0.03	Scroll speed. Zero freezes the layer.

Starting points

You want	Do this
Change the time of day	Rotate the scene Directional Light. With Sun Follows Main Light on, the disc, the glow and the light direction stay in agreement.
A clear daytime sky	Raise Zenith Color towards a saturated blue, keep Atmosphere Strength near 0.4, leave clouds off.
A sunset	Set Sky Gradient Strength to about 0.4, Direction to the compass bearing of the sun, and Gradient Color to a warm orange. Lower Horizon Sharpness so the warm band climbs higher.
Overcast	Clouds on, Coverage near 0.85, Softness near 0.3, Density 1.0, Cloud Color and Cloud Shadow both greyish and close to each other.
A stylised cel-shaded sky	Clouds on, Bands 3 to 5, Softness near 0.02, Saturation around 1.3.
Night	Darken Zenith and Horizon, drop Exposure to about 0.5, assign a star cubemap and lower Stars Threshold.

Performance

- The sky is one fullscreen pass with no texture reads except the optional star cubemap.
- Clouds and stars are shader keywords, not branches. With both switched off their code is not in the compiled shader at all.
- Cloud styles Round and Cotton walk a 3 by 3 cell neighbourhood, so they cost more than Standard, Feathered and Diagonal. On mobile prefer Standard.

- Stretch and Swirl each add a noise lookup pair. Leaving both at zero skips that work.

If something looks wrong

Symptom	Cause and fix
Sky is magenta	The project is not on URP, or the URP package is missing. The shader includes the URP shader library and cannot compile without it.
Sun does not move with the light	Sun Follows Main Light is off, or the scene has no light tagged as Main Directional Light in the Lighting window.
No stars	The Stars Cubemap slot is empty. Assign a star cubemap, then lower Stars Threshold.
Clouds do not move in the editor	Scene view animation is off. Enable Always Refresh in the Scene view toolbar, or enter Play mode.
Scene objects stay lit by the old sky	Environment Lighting Source is not set to Skybox, or lighting has not been regenerated.

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